

SEMP

IPS Solution

Digital Interlocking OBB

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

APPROVAL		
	Name and Role/Job Title	Esignature
Written by	V. Vasilev SEM	
Verified by	Ramy Al Ali QAM	
Approved by	E. Stemmer SEM – Generic Application	

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

LOG OF CHANGES			
Revision	Date	Author	Change Description
01 PD 01	2025-03-28	V. Vasilev	First proposal for review
01 PD 02	2025-09-25	V. Vasilev	Updated version aligned with the response to the Fourth Bid – SRR1 Proposal
01 RL	2025-10-02	V. Vasilev	Released version for SRR1

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

CONTENTS

1.	SCOPE	8
1.1	Identification	8
1.1.1	General	8
1.1.2	Solution Element of Solution	9
1.1.3	WBS managed as Work Packages	10
1.2	System overview	10
1.3	Key points	13
1.3.1	Milestones for Solution	13
1.3.2	Deliverables	14
1.4	Relation with other plans	14
1.5	Baselines of Software Configuration Items and Documentation Labels	15
1.5.1	Documentation labeling	15
1.6	Document version history	16
1.6.1	Edition 01	16
2.	REFERENCES AND TERMS	17
2.1	Referenced documents	17
2.2	Process basis	19
2.3	Standards and procedures	19
2.4	Terms, abbreviations and acronyms	21
2.4.1	Role abbreviations	21
2.4.2	General Terms, abbreviations and acronyms	22
2.4.3	Project Specific Terms, abbreviations and acronyms	23
3.	SOLUTION ENGINEERING AND DEVELOPMENT PROCESS	25
3.1	Organization of solution engineering	25
3.1.1	Solution core team	25
3.1.2	Solution team	26
3.1.3	External project roles	26
3.1.4	Configuration Control Board (CCB)	26
3.1.5	Subcontractors	27
3.2	System engineering supplies	27
3.3	Input data	27
3.4	Preconditions / Related solutions	27
3.5	Technical objectives	27
3.6	Detailed First Level WBS for Work Packages	28
3.7	Special need for training of system engineering team	28
3.8	Constraints / Risks	29
3.9	Work Contained	29
3.9.1	Work contained in Phase1	29
3.9.2	Work contained in Phase2	29

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.9.3	Work contained in Phase3	30
3.10	Work not contained	30
3.11	Responsibility Assignment Matrix	30
3.12	Tools and Facilities	30
3.12.1	Management and Documentation Tools.....	31
3.12.2	Code Building Tools.....	31
4.	PROCESS CONTROL.....	33
4.1	Control of Solution Development	33
4.1.1	Deviation.....	33
4.1.2	Metrics	33
4.2	Monitoring and Control of Solution Engineering Activities.....	35
4.2.1	Periodic Internal Reviews and Meetings.....	35
4.2.2	Milestone Reviews and Audits	35
4.2.3	Reporting	35
4.3	Critical Technologies.....	35
4.4	Risk and Opportunities Control	36
4.5	Corrective Action Process.....	36
4.6	Independent Evaluation and Verification.....	36
4.7	Estimate/Re-Estimate	36
4.8	Planning.....	36
4.9	Value Analysis and Design to Cost.....	36
4.10	Peer Reviews.....	36
5.	DEVELOPMENT SCENARIO AND TAILORING OF THE SYSTEM ENGINEERING PROCESS.....	37
5.1	Tailoring for this Solution (DDQS phases)	37
5.1.1	Orient the Solution	37
5.1.2	Design the Solution.....	37
5.1.3	Develop the Solution	37
5.1.4	Integrate and Verify the Solution	37
5.1.5	Validate the solution.....	37
5.2	Tailoring of Documentation Management	38
5.3	Tailoring of milestones	38
5.4	Tailoring of role assignments	38
5.4.1	Role of IVVQM	38
5.4.2	Role of Verifier (VER) according CENELEC EN 50126 and EN 50716.	38
5.4.3	Role of DM	38
5.5	Tailoring of the monitoring of project progress within an agile development environment.....	39
6.	SYSTEM ENGINEERING SUPPORT	40
6.1	Data and Configuration Management	40

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

6.2 Confidentiality 40

6.3 Export Control 40

6.4 Capitalization 40

6.5 Measurements and Analysis 40

6.6 Technical and Non-technical Requirements Management and Control 40

6.7 Validation of Requirements 40

6.8 Decision Analysis and Resolution 40

6.9 Subordinate Plans..... 41

6.10 Quality Assurance..... 41

6.11 Certification Support Activity 41

6.12 HSE Plan 41

APPENDIX A CRITERIA FOR THE SELECTION OF TECHNIQUES AND MEASURES

42

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

LIST OF TABLES

Table 1: Solution Identification.....	8
Table 2: Security / Maturity Level	8
Table 3: WBS managed as Work Packages	10
Table 4: Milestones Solution Element IPS Solution	13
Table 5: Subordinate Planning Documents.....	15
Table 6 : Referenced Documents	17
Table 7 : Standards and Procedures	19
Table 8 : Role abbreviations.....	21
Table 9 : Terms, Abbreviations and Acronyms	22
Table 10 : Terms, Abbreviations and Acronyms	23
Table 11: Solution Element Core team	25
Table 12: Solution Team	26
Table 13: External project roles	26
Table 14: CCB26	
Table 15: Detailed WBS	28
Table 16: Management and Documentation Tools	31
Table 17: GTS Control - Continuous Build.....	31
Table 18: Build	32
Table 19: Metrics.....	33
Table 20: Reviews and Meetings.....	35

LIST OF FIGURES

Figure 1: Solution Overview Generic Application Digital Interlocking OBB (DIL).....	9
Figure 2: Scope Split.....	11
Figure 3: System Architecture and Solutions.....	12
Figure 4: Develop and Qualify the Solution (DDQS).....	37
Figure 5: Interaction of solution and solution elements.....	37

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

1. SCOPE

The System Engineering Management Plan (SEMP) describes the organisation, work packages, milestones and application of the Management System DDQS process release number 2025-08 .

1.1 IDENTIFICATION

1.1.1 GENERAL

Table 1: Solution Identification

Project Name:	DSTW ÖBB Entwicklung
Project ID:	100025795 / ATRELDE91
Solution / System:	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
Solution Element:	Solution Element – IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0) ¹
Contract:	Digitales Stellwerk DSTW – Rahmenvereinbarung ²
Identification³:	IPS 40.0.0 ADM 40.0.0 MINIOMC 40.0.0 HW GENCAB SERV 1.0 HW GENCAB NETW 1.0
Customer:	Generic Application Digital Interlocking OBB (GA DIL OBB 1.0.0)
End Customer:	ÖBB
SIL Level:	NON-SIL / Non-safety solution ⁴

Table 2: Security / Maturity Level

Security Level:	SL-T 3, SL-C 3 ⁵
Maturity Level:	ML 3

¹ Please note that at this stage the focus of the development of the IPS Solution is on Phase 1 and this is indicated by 1.0.0 – for the phases see 3.9.1

² Please note that this SEMF Edition is based on the 4th Edition of the offer placed to ÖBB at 16th of June 2025

³ The identification refers to the key products/solutions included in the Solution Element - IPS Solution

⁴ For IPS and ADM only a validation report will be available

⁵ On the SL please note that ÖBB and also EULYNX clearly asks for SL3, an end-to-end implementation in Phase 1 will most likely NOT be possible. However, the setup and the describing documents shall be prepared for SL3. MiniOMC will not be classified as SL3 since it is based on OMC 3.0

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

This SEMP only deals with the scope of the price position Basic Virtualization (B-Vi) of the solution and contract mentioned in Table 1: Solution Identification.

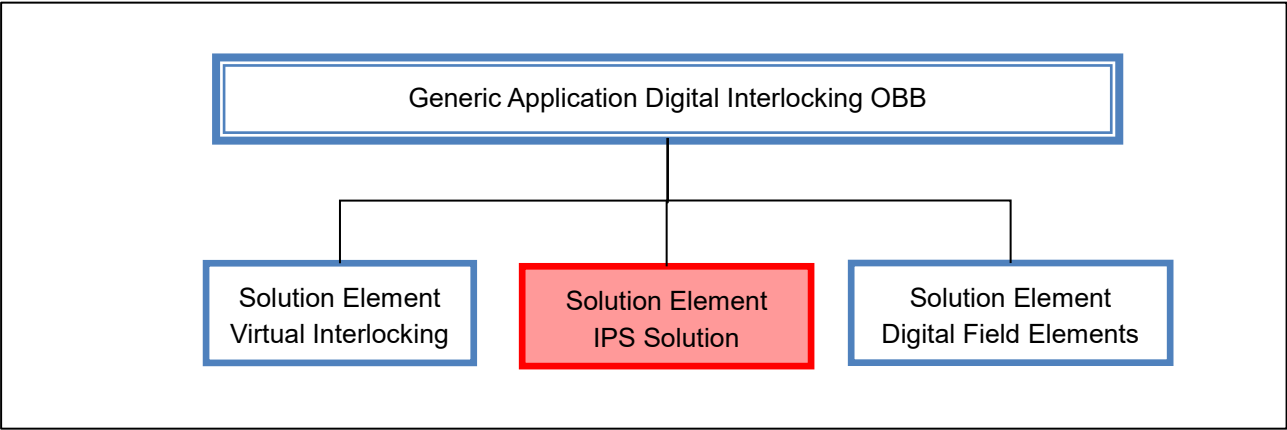
“Basic virtualization” (B-Vi) is the minimum technical requirement to be implemented upon the first delivery of a DIL including the implementation of all functions of the EULYNX specifications.

1.1.2 SOLUTION ELEMENT OF SOLUTION

The Solution Element IPS Solution Phase 1 defined in this document delivers an integrated software solution to the Solution Generic Application Digital Interlocking OBB 1.0.0 described in [SEMP DIL]. The dependencies to other Solution Elements and Work Packages are outlined in the individual Work Package Descriptions:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

Figure 1: Solution Overview Generic Application Digital Interlocking OBB (DIL)



NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

1.1.3 WBS MANAGED AS WORK PACKAGES

A detailed view of all WBS elements is listed in section 3.6.

Table 3: WBS managed as Work Packages

Solution:		ATRELDE91-E0				
WBS	Ident.	WPM	Initial Version	Target Version	Target date	WPD
ATRELDE91-E0.WP3.DEV.3.2.2	WP IPS	C. Ruggenthaler	IPS 30.0.0	IPS 40.0.0	Milestone FQR, see section 1.3.1	[WP Desc. IPS]
ATRELDE91-E0.WP3.DEV.3.2.2	WP ADM	Z. Ribic	ADM 30.0.0	ADM 40.0.0	Milestone FQR, see section 1.3.1	[WP Desc. ADM]
ATRELDE91-E0.WP3.DEV.3.2.2	WP IPS HW	A. Amon	N/A	HW GENCAB SERV 1.0 HW GENCAB NETW 1.0	Milestone FQR, see section 1.3.1	[WP Desc. IPS HW]
ATRELDE91-E0.WP3.DEV.3.2.2	WP MiniOMC	C. Ruggenthaler	N/A	MINIOMC 40.0.0	Milestone FQR, see section 1.3.1	[WP Desc. MiniOMC]

* Note: This SEMP describes the first product version for the first stations/installations; to fulfil the complete set of customer requirements two additional product versions are planned and will be described in future versions of the SEMP

1.2 SYSTEM OVERVIEW

For a comprehensive overview of the customer expectations about DIL, please refer to: [CRS] „OBB_RA-DSTW-D1_Leistungsbeschreibung“

The following is a summary of the purpose and scope:

1. **Development:** Creation of the new ELEKTRA interlocking generation based on the ÖBB specifications for the digital interlocking solution and new digital field elements
2. **Virtualization:** Potential virtualization of up to 287 interlockings (IxL) at two to three centralized computing centers, including geo-redundancy.

NOT EXPORT CONTROLLED

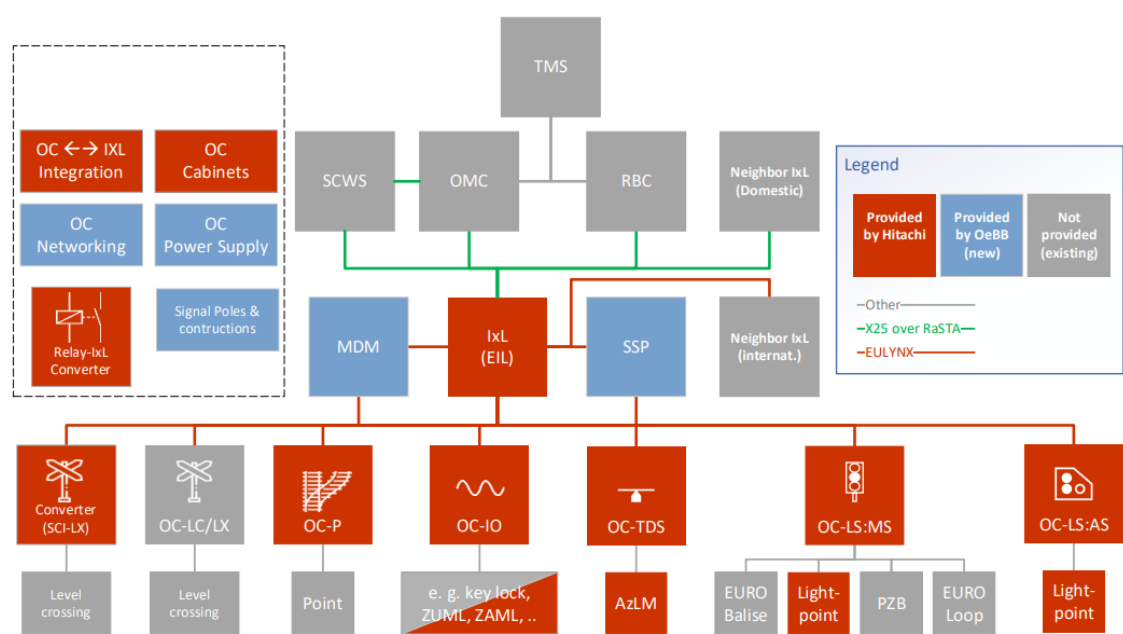
Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

- 3. **Reduction:** Decrease in the need for facilities/buildings for the interlocking core and maintenance requirements for field elements through virtualization and object controllers.,
- 4. **Standardization:** Addressing European standardization efforts by railway operators, known as the "EULYNX initiative."
- 5. **Cyber Security:** Implementation of cyber security measures based on IEC 62443, TS50701, and EULYNX BL 4, addressing the upcoming NIS2 directive and evolving cyber threat landscape.
- 6. **Roll-Out:** Enabling roll-out projects
- 7. **Service Infrastructure:** Establishing a service infrastructure
- 8. **Integration:** Overall integration of the interlocking solution into the ÖBB infrastructure.

Figure 2: Scope Split



For more information about used products see [PBS].

NOT EXPORT CONTROLLED

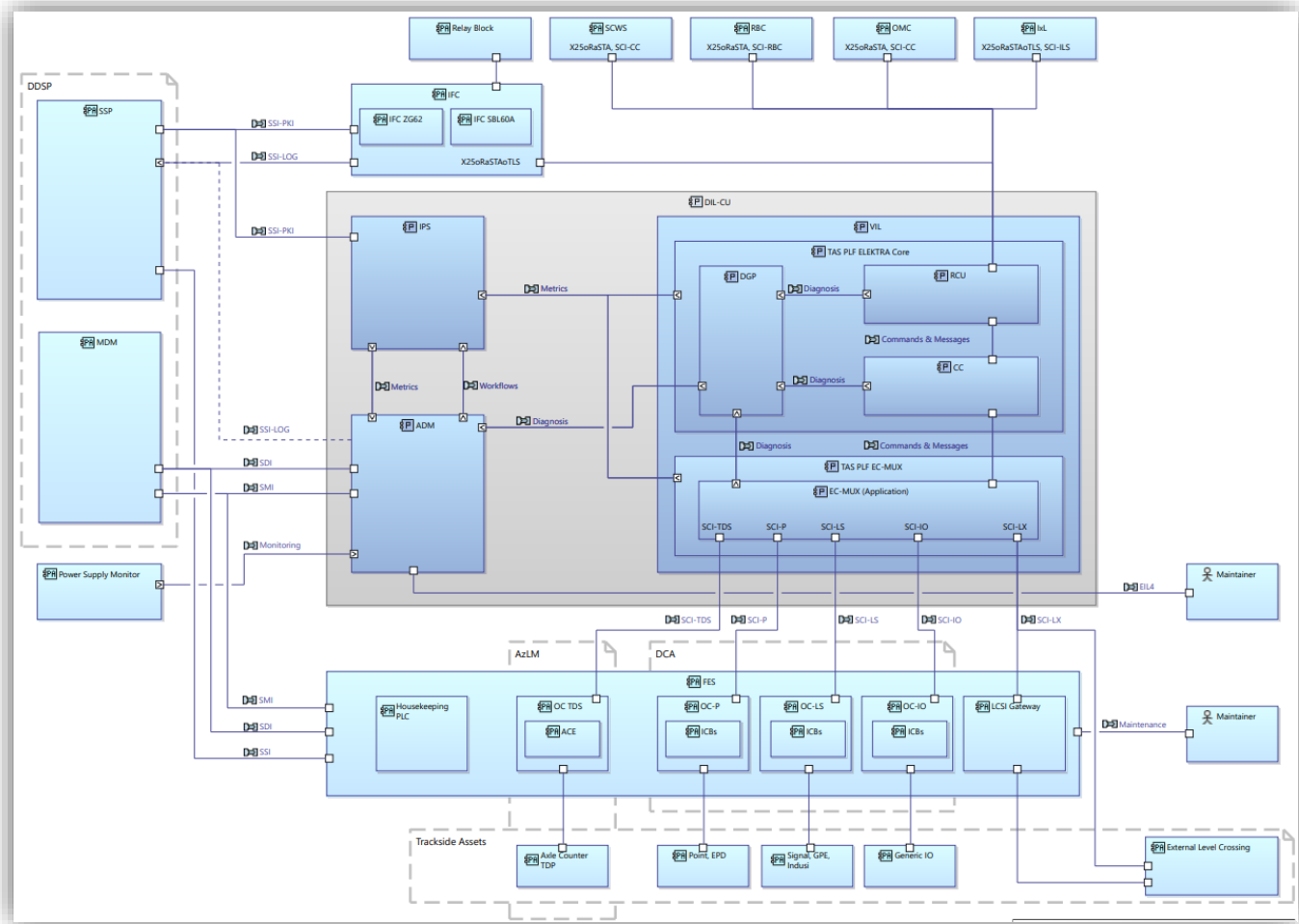
Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

The following figure illustrates the top-level system architecture. For a more comprehensive overview of the system and its architecture, please refer to [SSDD DIL], [SSDD IPS] and [SSDD ADM]

Figure 3: System Architecture and Solutions



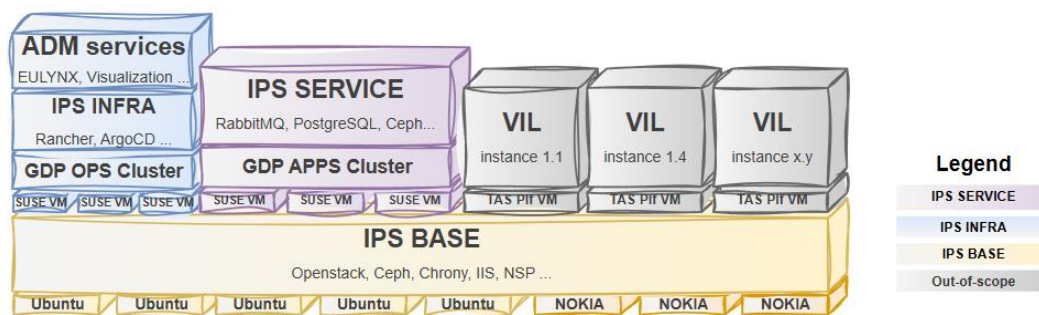
Within DIL, IPSS will provide the central virtualization platform including necessary administration workflows used by Virtual Interlocking (VIL) and Administration, Diagnosis & Monitoring (ADM) as depicted below.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved



IPS serves as the base virtualization platform to run and manage Virtual Machines (VMs) like VIL NMR applications.

Further, application and hardware specific workflows may be triggered and executed by ADM to manage, monitor, rollout and update the solutions including VIL, ADM, IPS and hardware configurations.

1.3 KEY POINTS

1.3.1 MILESTONES FOR SOLUTION

The following section outlines the DDQS milestones for the Solution Element IPS Solution – Phase 1. For details on customer and payment milestones, please refer to [PMP] and [SEMP DIL].

Table 4: Milestones Solution Element IPS Solution

Milestone	Target Date	Lead
Solution Orientation Review 1 (SOR1)	03.04.2025	SEM
Solution Orientation Review 2 (SOR2)	T0 ⁶ + 1 month	SEM
Solution Requirement Review 1 (SRR1)	03.10.2025	SEM
Solution Requirement Review 2 (SRR2)	T0 ⁶ & MS E03 ⁷ + 1 month	SEM
Solution Functional Review (SFR)	T0 ⁶ & MS E03 ⁷ + 4 months	SEM
Preliminary Design Review (PDR)	Agile approach	SEM
Critical Design Review (CDR)	Agile approach	SEM
Test Readiness Review (TRR1)	30.04.2027	IVM
Technical Qualification Review 1 (TQR1)	30.08.2027	IVM
Test Readiness Review 2 (TRR2)	30.08.2027	IVM
Technical Qualification Review 2 (TQR2)	TBD	SEM

⁶ All dates are under the assumption of a contract signature by end of December 2025 (=T0). For latest available dates see [IMP].

⁷ MS E03 Pflichtenheft Freigabe from Customer OBB defined in [PMP]. MS E03 is currently forecasted with April 2025. For latest available dates see [IMP].

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Final Qualification Review (FQR)	30.6.2028 ⁸	SEM
----------------------------------	------------------------	-----

- All dates are under the assumption of a contract signature by end of December 2025 (=T0)
- The milestones are related to the content of the Phase 1 of deliverables as included in the offer submitted to ÖBB in March 2025 – for details see section 3.9.1 Work contained in Phase1

1.3.2 DELIVERABLES

The deliverables for Solution Element IPS Solution include implementation, testing, verification, validation and integration of the new functionality as described in detail in the following Work Packages:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

1.4 RELATION WITH OTHER PLANS

See also Documentation Plan [DoP].

The SEMP describes the application of the Management System (Release from 2025-08) to the design, development and integration of the Solution Element IPS Solution Phase 1.

Title	Description	Ref.
<i>System Engineering Management Plan</i>	Focuses on the Solution Management of Solution Generic Application System including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[[SEMP DIL]]
<i>System Engineering Management Plan VIL</i>	Focuses on the Solution Management of Solution Element VIL including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[SEMP_VIL]
<i>System Engineering Management Plan DFE</i>	Focuses on the Solution Management of Solution Element DFE including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[SEMP SE DFE]
<i>Project Management Plan</i>	Focuses on the Management System including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project.	[PMP]

⁸ The overall target is to have a first station in Operation by end 2028 – therefore the FQR date for the Generic Application is set latest mid of 2028 overlapping with first Rollout activities and generation of Specific application

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Title	Description	Ref.
<i>Configuration Management Plan</i>	Provides details of the formal Configuration Management process.	[CMP]
<i>Quality Management Plan</i>	Describes requirements for all quality assurance measures and quality assurance activities to be carried out in the solution as contribution to the fulfilment of the project goals.	[QMP]
<i>RAM Plan</i>	Provides details of the tasks and activities related to Reliability, Availability and Maintainability of the Solution. Refer to Generic RAM Plan and two Project Specific RAM Plans.	[RAMP]
<i>Safety Plan</i>	Provides details of the tasks and activities of Solution safety management and Solution safety engineering required to identify, evaluate and eliminate or control hazards throughout the life cycle. Reference to Project Specific Safety Plan	[PSP]
<i>IV Plan</i>	Is the general overview how it is planned to manage and test the Generic Application with the three Solution Elements. The generic IVP and a project specific plan following the ÖBB requirements is provided	[IVP]
<i>VQ Plan</i>	Is the general overview how it is planned to manage and validate the Generic Application with the three Solution Elements. The generic VQP and a project specific plan following the ÖBB requirements is provided	[VQP]
<i>Security Management Plan</i>	Is the general overview how it is planned to address the security aspects in the Generic Application with the three Solution Elements. The generic SecMP and a project specific plan following the ÖBB requirements is provided	[SecMP]

Table 5: Subordinate Planning Documents

1.5 BASELINES OF SOFTWARE CONFIGURATION ITEMS AND DOCUMENTATION LABELS

1.5.1 DOCUMENTATION LABELING

For documentation available via DWF following labels are used as defined also in the Documentation Plan [DoP]

Details on the Product versions are given in the [CMP] and summarized in Table 3.

The following labels shall be applied to documents at the level of the Solution Element - IPS Solution

- “IPSS_OBB_1.0.0” for the documents which defines the first Baseline (Phase 1) of the IPS Solution

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

1.6 DOCUMENT VERSION HISTORY

1.6.1 EDITION 01

Initial edition.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

The relevant version of a document is referenced in the Documentation Plan [DoP]. Several planning documents are concurrent in progress until milestone SOR, therefore the edition 01 is referenced of Documentation Plan / Process Verification Report. Until milestone FQR the Documentation Plan / Process Verification Report for this Solution will be updated and tracked within the Version in SharePoint and DWF see [DoP].

Table 6 : Referenced Documents

Id	Nr./Title	Reference	Version
[CMP]	Configuration Management Plan ELEKTRA	3BU 15000 3338 QMAPA	11
[CMP_IPSS]	DSTW CMP	3BU 17300 1210 DUAPA	01PD01 (planned)
	CMP – IPS Generic	3BU 51799 0001 QMAPA	01
[CRS]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte	3BU 17300 0004 DSAPC	04
[DoP]	Documentation Plan (DoP)	3BU 17300 2000 ADAPA	01
[EngDB]	Engineering Dashboard (SharePoint)	100025795 ATRELDE91 DSTW OBB-IPSS EngDB	monthly
[IMP]	Integrated Master Schedule	Eng Master Schedule in SharePoint	latest
[IVP]	Integration and Verification Plan, ELEKTRA 2.x	3BU 15000 0654 DUAPA	03
	Project Specific Integration and Verification Plan (IVP)	3BU 17300 1206 DUAPC	01
[PBS]	Product Breakdown Structure	3BU 17300 1215 DPAPA	02
[PMP]	Project Management Plan	3BU 17300 1000 DPAPC	01PD02
[PSP]	Safety Plan DSTW	3BU 17300 1204 DUAPA	01
[QMP]	Quality Management Plan ELEKTRA	3BU 15000 5244 QMAPA	08PD01
	Quality Management Plan DSTW	3BU 17300 1202 QMAPC	01PD01 (planned)
[RAMP]	RAM Plan Generic	3BU 15000 1712 DUAPA	06
	RAM Plan DSTW	3BU 17300 1209 DUAPA	01
	RAM Plan IPS Solution SE	3BU 17300 2009 DUAPA	01PD01 (planned)

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Id	Nr./Title	Reference	Version
[RiskReport]	Risk Report	100025795_ATRELDE91-E0 DSTW ÖBB Entwicklung_Risk_List_20250930_ED02.xlsx	02
[RMP]	Requirements Management Plan (RMP) Generic Application Digital Interlocking OBB	3BU 17300 1207 DUAPA	01PD05
[RMP IPS]	Requirements Management Plan (RMP) Solution Element IPS Solution	3BU 17300 2010 DUAPA	01PD01
[SecMP]	Product Security Management Plan Elektra Project Specific Security Management Plan (SecMP)	3BU 15000 5300 DPAPA 3BU 17300 1213 DUAPA	03 01PD02
[SEMP DIL]	System Engineering Management Digital Interlocking OBB	3BU 17300 1200 DPAPA	02PD01
[SEMP SE DFE]	SEMP Solution Element - Digital Field Elements	3BU 62800 0000 DPAPA	01
[SEMP_VIL]	System Engineering Management Plan Solution Element Virtual Interlocking	3BU 15000 6955 DPAPA	01PD01
[SSDD ADM]	System / Subsystem Design Description (SSDD) – ADM	3BU_17300_2102_DSAPA	01PD01 (planned)
[SSDD DIL]	SSDD Digitales Stellwerk - Zentraleinheit	3BU_15000_1254_DSAPA	01PD01
[SSDD IPS]	System / Subsystem Design Description (SSDD) - IPS	3BU_17300_2052_DSAPA	01PD01
[SSS ADM]	System Subsystem Specification ADM- Subsystem, Digital Interlocking OBB	3BU_17300_2101_DTAPA	01PD02
[SSS DIL]	System Segment Specification (SSS) - Generic Application - Digital Interlocking OBB	3BU_17300_1251_DTAPA	01PD03
SSS IPS	System Subsystem Specification IPS-Subsystem, Digital Interlocking OBB	3BU_17300_2051_DTAPA	01PD02
[SSS SEC DIL]	System Segment Specification (SSS) - Cybersecurity DSTW ÖBB	3BU_17300_1252_DTAPA	01 (planned)

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Id	Nr./Title	Reference	Version
[ToolClassSW]	Software Development, Tool Classification, EN 50128	3BU 15000 5135 AAAPA⁹	latest
[ToolClass]	Thales Austria Tool Classification EN50129	3BU 15000 5135 190EN	latest
[VQP]	Validation and Qualification Plan, ELEKTRA 2.x Project Specific Validation and Qualification Plan (VQP)	3BU 15000 0655 DUAPA 3BU 17300 1205 DUAPA	03 01
[WP Desc. IPS]	Work Package Description (WPD) IPS for DIL - IPS Solution	https://e2.at.gts/confluence/x/HbHVCg 3BU 17300 2050 ADAPA	Latest 01
[WP Desc. ADM]	Work Package Description (WPD) ADM for DIL - IPS Solution	https://e2.at.gts/confluence/x/perVCg 3BU 17300 2100 ADAPA	Latest 01
[WP Desc. MiniOMC]	Work Package Description (WPD) MiniOMC for DIL - IPS Solution	https://e2.at.gts/confluence/x/eK8bCw 3BU 17300 2150 ADAPA	Latest 01
[WP Desc. IPS HW]	Work Package Description (WPD) IPS HW for DIL - IPS Solution	https://e2.at.gts/confluence/x/yQUTD	Latest

2.2 PROCESS BASIS

The project is set up with the Management System, release number 2025-08 as a process basis. Changes in this process basis (e.g. adaptation to a new release of Management System) have to be analysed, especially regarding the impact on already existing work products, and have to be documented and approved.

2.3 STANDARDS AND PROCEDURES

Table 7 : Standards and Procedures

Nr./Title	Publishing Date/Reference	Version/Revision
[S1] IRIS Certification rules:2020 based on ISO/TS 22163:2017	2017	n/a
[S2] EN 50716:2023 – Railway application – Requirements for software development	2023	n/a

⁹ This document will need to be updated according to the successor [S2] EN 50716:2023 – Railway application – Requirements for software development of EN 50128

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Nr./Title	Publishing Date/ Reference	Version/ Revision
[S3] EN 50129:2018 + AC:2019 Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling	2018/2019	n/a
[S4] ISO 9001	2015	n/a
[S5] EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process	2017	n/a
[S6] EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety	2017	n/a
[S7] Management System Process Manage Risks and Opportunities	Part of 87200984-00-PRJ-GRP-EN (Chapter 8.6)	2025-08
[S8] Management System Process Manage Project	87200984-00-PRJ-GRP-EN	2025-08
[S9] Minimum Requirements for the configuration management implementation	Part of 87100077-CFG-GRP	007
[S10] Management System Process 6.1 DDQS-SW - Design develop and qualify the Software solution	Part of 87202125-00-DDQ-GRP-EN	2025-08
[S11] Management System Process 6.1 DDQS-HW - Design develop and qualify the Hardware solution	Part of 87202126-00-DDQ-GRP-EN	2025-08
[S12] Management System Process Validate the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S13] Management System Process Integrate and Verify the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S14] Management System Process Design the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S15] Management System Process 9.7 CFG - Manage Configuration	87202424-00-CFG-GRP-EN	2025-08
[S16] Management System Process 9.3 Control Export	87299999-00-EXP-GRP-EN	2025-08
[S17] Management System Guarantee quality and manage continuous improvement	87200214-00-QCI-GRP-EN	2025-08
[S18] Protection of Group Information	87201725-GOV-GRP-EN	005
[S19] Personal Data Protection Policy	87212828-GOV-GRP-EN	001
[S20] Project Life Cycle	83530275-DDQ-AUT-EN	005
[S21] Document Management Rules	83530061-DOC-AUT-EN	013
[S22] Milestone Template Solutions	83530019-DDQ-AUT-EN	013

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Nr./Title	Publishing Date/ Reference	Version/ Revision
[S23] Document List and Review Matrix for engineering projects	83530224-DOC-AUT-EN	010
[S24] HSE Instruction	83530048-HSE-AUT-DE	020
[S25] Agile Definition SMP	83530312-DDQ-AUT-EN	002
[S26] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-3-3: System security requirements and security levels, Edition 1, 09 2013	2013	01
[S27] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-1: Secure product development life-cycle requirements, Edition 1, 01 2018	2018	01
[S28] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-2: Technical security requirements for IACS components, Edition 1, 04 2019	2019	01
[S29] EN 50159, + A1:2020 Railway applications – Communication, signalling and processing systems – Safety-related communication in closed transmission systems	2020	n/a
[S30] Template of DoX Templates	83530313-DDQ-AUT-EN	002

2.4 TERMS, ABBREVIATIONS AND ACRONYMS

2.4.1 ROLE ABBREVIATIONS

Table 8 : Role abbreviations

ARC	Architect
CM	Configuration Manager
DM	Document Manager
HoD	Head of Discipline
IVM	Integration and Verification Manager
IVVQM	Integration, Verification, Validation and Qualification Manager
PDA	Project Design Authority
PLA	Product Line Architect
PLM	Product Line Manager
PSM	Project Security Manager
QAM	Quality Assurance Manager
RAM-AM	RAM Assurance Manager
RE	Requirements Engineer

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

RM	Requirements Manager
TM	Test Manager
SAM	Safety Assurance Manager
SecAM	Security Assurance Manager
SEM	Solution Engineering Manager
SM	Senior Management
VAL	Validator
VER	Verifier
WPM	Work Package Manager

2.4.2 GENERAL TERMS, ABBREVIATIONS AND ACRONYMS

Table 9 : Terms, Abbreviations and Acronyms

BCWS	Budgeted Cost of Work Scheduled
CBB	Contract Budget Baseline
CCB	Configuration Control Board
CDR	Critical Design Review
CENELEC	European Committee for Electrotechnical Standardization
CHARM	Change and Request Management
CR	Change Request
DDQS	Design, Develop and Qualify the Solution
EDM	ELEKTRA Delivery Model
EV	Earned Value
FEA-ID	Feature Identifier from Feature List
FQR	Final Qualification Review
GEN	Generic
HW	Hardware
IFS	Interfaces
IVV	Integration, Verification and Validation
KOM	Kick of Meeting
M	Mandatory
MPP	Manage Product Policy
N/A	Not applicable
O	Optional

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

PCM	Primavera Core Model
PCR	Problem Change Request
PDR	Preliminary Design Review
PEP	Personnel Efforts Planning
PiO	Put in Operation
PMP	Project Management Plan
QMP	Quality Management Plan
RAMS	Reliability, Availability, Maintainability and Safety
SE	Solution Element
SEMP	System Engineering Management Plan
SFR	Solution Functional Review
SIL	Safety Integrity Level
SOR	Solution Orientation Review
SRR	Solution Requirement Review
SW	Software
TQR	Technical Qualification Review
TRR	Test Readiness Review
WBS	Work Breakdown Structure
WP	Work Package
WPD	Work Package Description

2.4.3 PROJECT SPECIFIC TERMS, ABBREVIATIONS AND ACRONYMS

Table 10 : Terms, Abbreviations and Acronyms

ADM	Administration Diagnostic Monitoring
AzLM	Multi Section Axle Counter (one type of TDS)
BI	Basic Integrity
CFI	Customer Furnished Item
DCA	Distributed Controller Architecture
DDSP	Daten Diagnose Service Plattform (CFI ÖBB)
DFE	Digital Field Elements
DGP	Diagnosis Processor (ELEKTRA)
DIL	Digital Interlocking = DSTW
DSTW	Digitales Stellwerk = DIL

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

DSTW-ZE	Digitales Stellwerk Zentraleinheit = EIL
ECMUX	Element Controller MultipleXer
EIL	Electronic Interlocking System according to EULYNX – equals to DSTW-ZE
EKSA	Eisenbahnkreuzungssicherungsanlage = LCSi (CFI ÖBB)
EULYNX	European Initiative to Linking Interlocking Systems
FES	Field Element Station
IEC	International Electrotechnical Commission
GA DIL ÖBB	Generic Application Digital Interlocking ÖBB
IPS	Integrated Platform Services
IPSS	IPS Solution Element
LCSi	Level Crossing Safety Installation = EKSA
LX	Level Crossing
MDM	EULYNX, Maintenance and Data Management
ÖBB	Oesterreichische Bundes Bahn (Austrian Railway Operator, ÖBB)
OC	Object Controller
OMC	Operation Management Center = BFZ (Betriebsführungszentrale)
RaSTA	Rail Safe Transport Application (safety protocol for railway signalling systems)
RBC	Radio Block Center (CFI ÖBB)
RCU	Remote Control Unit
SCI	Standard Communication Interface (EULYNX interface for process data information)
SCWS	Signal Controlled Warning System
SDI	Standard Diagnostic Interface (EULYNX interface for diagnostics)
SiDiNet	SicherungsDienstNetzwerk (CFI ÖBB)
SMI	Standard Maintenance Interface (EULYNX interface for maintenance)
Solution	A consistent set of systems, equipment and services provided to the Customer to meet a specific requirement
Solution Element	A system, a sub-system, a product / equipment, or a service, a software component, a hardware component
SSI	Standard Security Interface (EULYNX interface for security)
SSP	Security Services Platform (CFI ÖBB)
TDS	Train detection system
VIL	Virtual Interlocking
VSTW	Virtuelles Stellwerk = VIL

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3. SOLUTION ENGINEERING AND DEVELOPMENT PROCESS

The Solution will be developed according to Hitachi Rail GTS Austria Management System. The Management System version valid at the start of the Solution is 2025-01.

As the Management System continues to improve, it is important to note that for newly created documents within this project, the appropriate and current templates must be used in accordance with the latest Management System guidelines. For maintenance and extension of existing documents the previously used template may remain.

3.1 ORGANIZATION OF SOLUTION ENGINEERING

The description of the roles mentioned below can be found on the Management System Portal.

3.1.1 SOLUTION CORE TEAM

Table 11: Solution Element Core team

Role	Resource	Deputy Resource
PDA	M. Eichberger	
PLM	R. Holz knecht	
SEM	V.Vasilev	
IVM	M. Siska	
ARC	M. Franzl	
VER	see **) comment below	
SAM	S. Thoma	
QAM	R. Al Ali	A. Haslinger
VAL	M. Gronau	
RM	N. Karmatska	S. Bogilovic
WPM IPS / WPM MiniOMC	C. Ruggenthaler	
WPM ADM	Z. Ribic	
WPM IPS HW	A. Amon	

**) For details please refer to chapter 5.4.2 Role of Verifier (VER) according CENELEC EN 50126 and EN 50716

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.1.2 SOLUTION TEAM

Table 12: Solution Team

Role	Resource	Deputy Resource
CM	C. Ruggenthaler	
PSM	S. Cako	
SecAM	A. Willenberg	
DM	see *) comment below	n/a
RAM-AM	F. Moser	M. Scheuer
Integration	n/a	n/a

*) The role DM is not executed by a single person. For details please refer to chapter 5.4.3 Role of DM.

3.1.3 EXTERNAL PROJECT ROLES

Table 13: External project roles

Role	Resource	Deputy Resource
Safety Assessor	Arsenal Race	n/a

3.1.4 CONFIGURATION CONTROL BOARD (CCB)

Table 14: CCB

Role	Resource	Comment	M/O
PLM	see Table 11	Budget, schedule, scope	M
PM	see Table 11		M
SEM	see Table 11		M
PDA	see Table 11		M
SAM	see Table 11		M
IVM	see Table 11		O
VAL	see Table 11		M
QAM	see Table 11		M
Head of Engineering	M. Vana	for priority conflicts regarding resources in E	O

Other project team members are invited to the CCB as needed.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.1.5 SUBCONTRACTORS

IPS Solution have contracted Nokia Solutions and Networks Österreich GmbH to help with the Network Support services.

3.2 SYSTEM ENGINEERING SUPPLIES

Not applicable in this solution.

3.3 INPUT DATA

For the input data related to the Solution Element IPS Solution please refer to [SSS DIL], [SSS SEC DIL], [SSDD DIL], [SEMP DIL] and the following Work Packages:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

3.4 PRECONDITIONS / RELATED SOLUTIONS

For overview of the preconditions related to the Solution Element IPS Solution (IPSS) please refer to section 7.1, WP dependencies > Preconditions, of the following Work Packages:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

3.5 TECHNICAL OBJECTIVES

The requirements basis for the Generic Application is the [CRS] - DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte

NOT EXPORT CONTROLLED

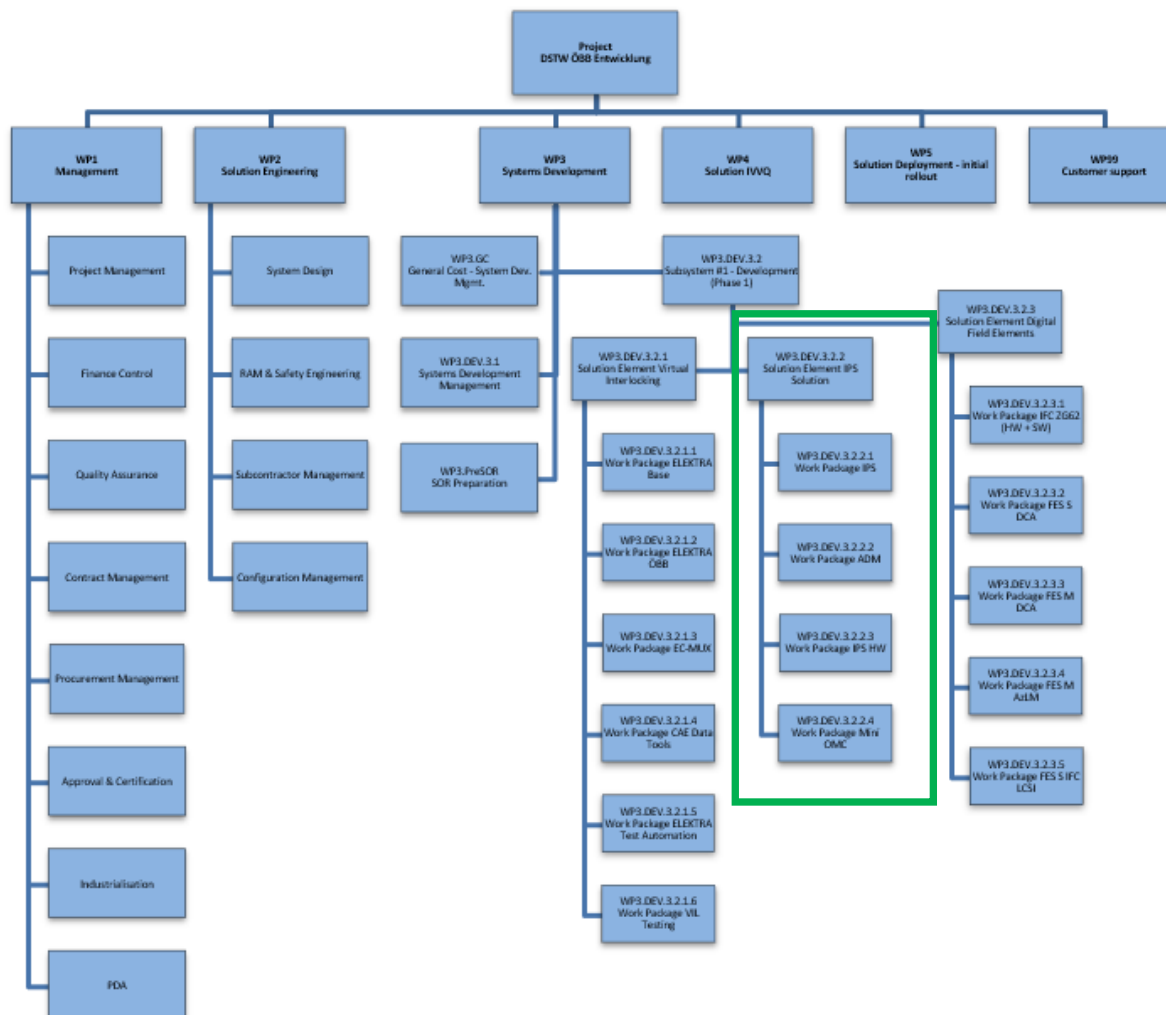
Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.6 DETAILED FIRST LEVEL WBS FOR WORK PACKAGES

Table 15: Excerpt of the Detailed WBS



3.7 SPECIAL NEED FOR TRAINING OF SYSTEM ENGINEERING TEAM

A training plan will be established, focusing on the following areas:

- SEM/WPM and IVM certification
- Introduction to new technologies (servers) and interfaces (EULYNX)
- Safety principles of DIL
- RAM Design Rules of DIL
- New concepts developed in the project, such as IPS background for all team members

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.8 CONSTRAINTS / RISKS

Constraints which are not already considered in the [RiskReport] are listed here.

- None.

3.9 WORK CONTAINED

This section provides an overview of the work contained in the solution. Detailed descriptions of the work involved, including the scope of work packages for the IPS Solution Element, covering implementation, testing, verification, validation, and integration of new functionalities, are outlined in the following Work Packages:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

Work Contained overview:

- Engineering of the scope mentioned in section 1.2 according to the applying parts of the relevant standards listed in section 2.3.
- Delivery of the baseline documentation.
- Delivery of the items listed in section 1.3.2. - Deliverables.

3.9.1 WORK CONTAINED IN PHASE1

- Hot-Hot Redundancy 2x2oo2 for ELEKTRA Core and 1oo2 for ECMUX based on TAS Platform 2.6.x and IPS-Solution
- "Static" Server Virtualization for the first stations based on OMC-3.0 solution. (Server Hardware provided by Hitachi - Network components and cabinets provided either by ÖBB or Hitachi)
Relatively static Setup- und Administration of the system
- Development and Qualification of a Server and Network Cabinet based on OMC 3.0 (including new Server Generation, new PDU, etc.
- Basic Diagnosis and Administration via ADM
- CMOS data recovery
- Initial lab and test environment
- Test automatization
- MiniOMC (OMC3 as a test tool) to enable VIL component & system tests
- Cybersecurity implementation package 1 -> (Security Level 2 included in Security Case)
ÖBB requests to be considered
- early ability to test end-end security
 - a. IPS feature set Phase 1
 - * key generation (private keys)
 - * SSI catalogue (proxy, NAT, SCI, SDI for testing purposes)

3.9.2 WORK CONTAINED IN PHASE2

- Corrections if any
- Additional CMOS data recovery in case of planned maintenance

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

- Flexible Setup- and Administration via ADM and DDSP
- Full diagnosis and maintenance functions
- Full Cybersecurity measures
- Lab and test environment extensions
- DDSP (CFI) full Diagnosis and Security
- Maintenance mode for field elements (Betrieb mit tlw. außer Betrieb befindlicher Außenanlage)

3.9.3 WORK CONTAINED IN PHASE3

- Corrections if any
- SCI-ILS (neighbor VSTW, neighbor country)
- DDSP Maintenance features
- Self check feature for OCs and DIL
- Software activation via OMC

3.10 WORK NOT CONTAINED

While the work contained is already stated in section 3.9, the following topics are explicitly mentioned as not part of this solution in order to sharpen the responsibility:

- Execution of any field tests on site
- Execution of customer acceptance tests
- Coordination and execution of Factory Acceptance Tests (Customer lab-/integration-tests)
- Installation and put into operation at customer site
- Additional features as offered to ÖBB like Geo-redundancy, E-Shu
- The integration of third-party OCs – however the integration activity is a clear request of the customer and can be requested by a separated purchase order

More comprehensive overview of the work not contained is outlined in the following Work Packages:

- [WP Desc. IPS]
- [WP Desc. ADM]
- [WP Desc. MiniOMC]
- [WP Desc. IPS HW]

3.11 RESPONSIBILITY ASSIGNMENT MATRIX

The responsibilities of the roles at a Milestone Review can be seen in Milestone Template Solutions [S22].

3.12 TOOLS AND FACILITIES

The following subsections provide a rough overview of the project tool setup. For version information and tool classification see document [ToolClassSW].

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.12.1 MANAGEMENT AND DOCUMENTATION TOOLS

Table 16: Management and Documentation Tools

Tool Category	Tool	Note
Project Management	Primavera PCM	
Task Management	Jira iTask	
Requirements Management	DOORS/RMF, Polarion	
Management Traceability	DOORS/RMF, Polarion	
Configuration Management	ClearCase Git	
Change Management	CHARM	
Documentation Management	ClearCase DWF SharePoint	
Documentation Generation	DOORS/RMF Polarion MS Office	
IVV Management	Polarion MS Excel	

3.12.2 CODE BUILDING TOOLS

Details regarding the specified tools are available in the [CMP_IPSS] document

3.12.2.1 GTS CONTROL - CONTINUOUS BUILD

Table 17: GTS Control - Continuous Build

Name	Version	App	Rationale	User max/Concurrent	Reference to validation report
Hudson + Subversion		☐			
Hudson + Clear Case		☐			
Maven		☒			
SonarQube		☒			
Jenkins + Bitbucket		☒			

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

3.12.2.2 GTS AT BUILD

Table 18: Build

Name	Version	App	Rationale	User max/Concurrent	Reference to validation report
Ansible	7.7.0	IPS ☒			
Ant		☐			
Bash		IPS ☐			
GitLab	v17.8.5	IPS ☐	OMC IPS – Usage in DSTW needs to be defined / aligned		
Make		☐			
Maven	3.3.9	☒	Upgrade under discussion		
Jenkins		☒			
Terraform		IPS ☐			

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

4. PROCESS CONTROL

4.1 CONTROL OF SOLUTION DEVELOPMENT

The solution element IPS assigned responsibilities are listed in

Table 3.

4.1.1 DEVIATION

The development of a Solution has to be planned and executed according to the DDQS Processes of Management System. Deviation to the description in Project Life Cycle [S20] is described in section 5.

4.1.2 METRICS

The used metrics for control of the engineering process are identified in the Engineering Dashboard [EngDB].

The tailoring for this dashboard is listed below:

Table 19: Metrics

	Sheet	Tailoring
☒	Actions & Decisions	Used - no deviation.
☐	Stability of Team Velocity	Not used. See Note 5 below.
☐	Visibility Point Burnup	Not used. See Note 5 below.
☐	Code Quality	Not used.
☐	Code Reuse	Not used.
☐	Defect management	Not used.
☒	Documentation	Used – only System Level documents reported
☒	Finance	Used. See Note 1 below.
☐	Interface testing progress	Not used.
☐	IVV Test Writing	Not used.
☒	Master Plan	Used - no deviation.
☒	Milestones	Used - no deviation.
☒	OBS	Used - no deviation.
☒	PCR (Problem Change Request)	Used. See Note 2 below.
☒	Quality	Used. Data provided by QAM.
☐	RAMS	Optional. Data provided by RAM-AM and SAM.
☐	Release content progress	Optional. Not only features are tracked, but also legacy PCRs which are part of the scope as defined in section 0.
☐	Remaining work burndown	Not used.
☒	Requirement Analysis	Used. See Note 3 below.
☐	Requirement Flowdown	Not used.
☐	Sizing & critical performance	Not used.
☒	Technical Risks	Used - no deviation.
☐	Validate & qualify test progress	Not used.
☒	Verification testing progress	Used. See Note 4 below.
☒	WBS	Used - no deviation.
☒	Workload	Used - no deviation.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Note 1: ⓘ Guideline:

- a) "BCWS Planned Spend" is the monthly planned costs of work. This value can be determined from PCM. It is the defined/updated at the time the CBB is defined or updated.
- b) "Forecast" is the spending forecast from the last month with actuals to the end of the project (before this period it has to be empty).
- c) "EV" is not used.

Note 2: ⓘ Guideline:

This metric deals with PCRs found in the phase "Integrate and Verify the Solution". Hence no baseline can be given ahead. The "Raised Forecast" may be predicted after a few months of this phase from past data.

Note 3: ⓘ Guideline:

Only the table "requirements chart data" is filled out, were the

- d) "Total number of Req.": is the sum of all approved requirements at the time of the reporting. The reporting is done every month
- e) "Baseline": a new baseline is created, when a new release of the SSS is done and a new release label is generated in DOORS
- f) "Analysis forecast": specifies the sum of current and new requirements.
- g) Either a value in "Baseline" or in the "Analyses forecast" is filled out except in the current month, where both fields are filled out.

Additionally, to be filled out are:

- "Analyzed Req.": sum of all requirements, which are analyzed and not changed
- "Req. to be clarified": sum of requirements, which are currently not approved, new or changed, may not be the sum of the below attributes
- "Obsolete Req.": sum of requirements, which will be set to descoped
- "Modified Req.": sum of changed requirements
- "New Req.": sum of new requirements

Requirements characterization can be generated via DOORS, via the menu "TRSS-AT" -> "Simple Functions" -> "KPI" -> "Input for Req. analysis tab".

Note 4: ⓘ Guideline:

Field "Req. conformity (%)" may be used as a percentage of tested and verified items of the defined scope

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

Note 5: ⓘ Guideline:

The monitoring of project progress within an agile development environment requires a set of metrics defined in the Agile Definition SMP [S25] document sections 9.2.1 'Stability of Team Velocity' and 9.2.2 'Visibility Point Burnup' and is expected by senior management to be displayed during the PPPRs.

Furthermore, the tailoring of the monitoring of project progress within an agile development environment in chapter 5.5 must be taken into consideration to get an adequate information basis for the KPIs.

4.2 MONITORING AND CONTROL OF SOLUTION ENGINEERING ACTIVITIES

4.2.1 PERIODIC INTERNAL REVIEWS AND MEETINGS

Table 20: Reviews and Meetings

Meeting	Responsible	Attendees	Frequency	Objectives
periodically Solution Core Team Meeting	SEM	"Solution core team", ARC, WPM(s), RM Only on demand: CM, IPS Team Lead	weekly	Project status and action points.
Risk and opportunity review	SEM	PM, SEM, PLM, PDA, IVM, SAM, ARC, QAM, WPMs, IPS Team Lead	Quarterly or on demand	
Resource review	SEM	PM, SEM, HoD, IPS Team Lead	monthly	Check utilisation and additional needs of solution team.
Cost tracking	Project controller	PM, SEM, HoD, IPS Team Lead	monthly	Report of costs and progress to Financial Control Director.
Periodically Performance Project Review (PPPR)	SEM	PM, SEM, QAM	monthly	Report of costs and progress, KPI dashboard, Progress Matrix, required decisions from SM to HoD Project Engineering, HoD Engineering

4.2.2 MILESTONE REVIEWS AND AUDITS

Milestone reviews are done according to Milestone Template Solutions [S22]. Quality Audits will be conducted through the QAM – see [QMP]. Safety Audits will be conducted through the SAM – see [PSP].

4.2.3 REPORTING

Reporting is done via the Engineering Dashboard [EngDB].

4.3 CRITICAL TECHNOLOGIES

Not applicable in this solution.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

4.4 RISK AND OPPORTUNITIES CONTROL

The Risk and Opportunities Control in this Solution will follow the Management System Process Manage Risks and Opportunities [S7] and is monitored in the [RiskReport].

4.5 CORRECTIVE ACTION PROCESS

Information about the Corrective Action Process can be found in the [QMP].

4.6 INDEPENDENT EVALUATION AND VERIFICATION

Independent evaluation and verification will be done by

1. the verifier and domain experts upon instruction of the verifier (VER);
2. the validator (VAL);
3. the safety assurance manager (SAM).

4.7 ESTIMATE/RE-ESTIMATE

The initial estimation is either an input for the phase “Orientation the Solution” or is done or verified during this phase. Re-estimation of remaining work is done during the progress of working packages typically on a monthly period. The method used for estimation is mainly “Expert estimations” based on past experience, educated assumptions and risk considerations.

4.8 PLANNING

Planning is done in the tool PCM and can be found there with the identification listed in Table 1.

4.9 VALUE ANALYSIS AND DESIGN TO COST

Value Analysis and Design to Cost is done by the Engineering Dashboard [EngDB].

4.10 PEER REVIEWS

The Peer Reviews in this Solution will follow the Minimum Requirements for the configuration management implementation [S9]. An overview of mandatory Documents and Reviewer has to be done according to Document List and Review Matrix for engineering projects [S23].

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

5. DEVELOPMENT SCENARIO AND TAILORING OF THE SYSTEM ENGINEERING PROCESS

This development process follows the initialisation by Management System process Manage Product Policy (MPP) as described in Project Life Cycle [S20].

Figure 4: Develop and Qualify the Solution (DDQS)

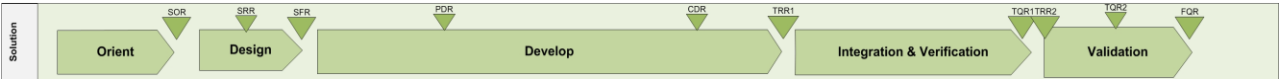
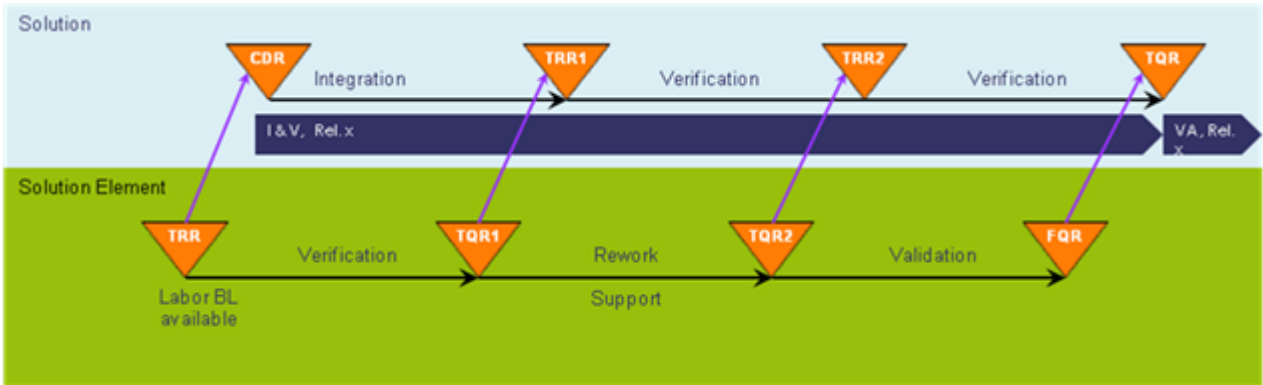


Figure 5: Interaction of solution and solution elements



5.1 TAILORING FOR THIS SOLUTION (DDQS PHASES)

5.1.1 ORIENT THE SOLUTION

No other deviation to the description in Project Life Cycle [S20].

5.1.2 DESIGN THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.3 DEVELOP THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.4 INTEGRATE AND VERIFY THE SOLUTION

No deviation to the description in Project Life Cycle [S20].

5.1.5 VALIDATE THE SOLUTION

The Assessment Certification according to CENELEC is needed for this solution. No other deviation to the description in Project Life Cycle [S20].

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

5.2 TAILORING OF DOCUMENTATION MANAGEMENT

The tailoring of the Product Line Engineering (PLE) is used as described in Project Life Cycle [S20].

The naming convention for the name “Firstname Lastname” of the author, verified by, approver of a document can also be “F. Lastname”.

No other deviation to the description in Document Management Rules [S21].

5.3 TAILORING OF MILESTONES

There are no deviations from the description in Project Life Cycle [S20] and Agile Definition [S23]. Any exception(s) are noted in the relevant sections of this document.

5.4 TAILORING OF ROLE ASSIGNMENTS

5.4.1 ROLE OF IVVQM

The role IVVQM is not executed by a single person. The major IVVQM activities are distributed to other roles:

- Integration: is made by IVM
- Verification: is made by IVM
- Validation: is made by VAL
- Qualification: is made by VAL (in case of HW development)

5.4.2 ROLE OF VERIFIER (VER) ACCORDING CENELEC EN 50126 AND EN 50716

The verification tasks will be handled according to CENELEC [S2] EN 50716:2023 – Railway application – Requirements for software development. The review of all documents is performed through reviews by the roles defined in the Document List and Review Matrix for engineering projects [S23]. The role VER is not executed by a single person. The responsibilities are defined in Table B.6 in [S2] and Table G.3 in [S6] (EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety) :

- Process Verification: is made by QAM
- Final document Verification: is made by SEM or a person assigned by the SEM within the [DoP]
- Software/Code Verification: Sprint peer reviews
- Deliverables Verification: is made by IVM
- Enabling systems Verification: is made by CM
- Lifecycle phases Verification: is made by QAM
- Collection of Verification results in Verification-report: is made by IVM or a person assigned by the SEM within the [DoP]

5.4.3 ROLE OF DM

The role DM is not executed by a single person. The major DM activities are distributed to other roles and anticipated in the estimation phase:

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

- new ANV: can be obtained by SEM
- Document Index: is made by CM or SEM
- Formal check at release of a document: is made by QAM
- Other DM activities: will be decided at point in time

5.5 TAILORING OF THE MONITORING OF PROJECT PROGRESS WITHIN AN AGILE DEVELOPMENT ENVIRONMENT

There are no deviations from the description in Project Life Cycle [S20] and Agile Definition [S23]. Any exception(s) are noted in the relevant sections of this document.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

6. SYSTEM ENGINEERING SUPPORT

6.1 DATA AND CONFIGURATION MANAGEMENT

Data and Configuration Management is carried out according to Management System Process 9.7 CFG - Manage Configuration [S15], Document Management Rules [S21] and the specific Configuration Management Plan [CMP].

6.2 CONFIDENTIALITY

The rules as listed in Protection of Group Information [S18] apply.

6.3 EXPORT CONTROL

N/A

6.4 CAPITALIZATION

Capitalization is handled in the context of the related Manage Product Policy process.

6.5 MEASUREMENTS AND ANALYSIS

See section 4.

6.6 TECHNICAL AND NON-TECHNICAL REQUIREMENTS MANAGEMENT AND CONTROL

For Requirement Management the tools DOORS and Polarion are used. The Requirements Manager ensures that requirements are

- uniquely identified
- unambiguous, testable and consistent
- traced from origin through all systems
- have all changes managed

The Requirements Manager manages changes throughout the life of the project. If requirements change, the impact will be reflected in the project plans, work products, and activities. Any inconsistencies between these customer- and/or system requirements and project plans, work products, and activities will be identified and resolved.

6.7 VALIDATION OF REQUIREMENTS

The validation is done in accordance with the Validation Qualification plan as listed in Documentation Plan [DoP].

6.8 DECISION ANALYSIS AND RESOLUTION

Decision analysis and resolutions are part of design and review processes. The results are documented in design documents, review reports and in this document.

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

6.9 SUBORDINATE PLANS

Subordinate plans according to Table 3: WBS managed as Work Packages are listed in Documentation Plan [DoP].

6.10 QUALITY ASSURANCE

The Quality Assurance in this Solution will follow Management System Guarantee quality and manage continuous improvement [S17]. Additional information can be found in the Quality Management plans as listed in Documentation Plan [DoP].

6.11 CERTIFICATION SUPPORT ACTIVITY

See Safety Management plans as listed in Documentation Plan [DoP].

6.12 HSE PLAN

The local requirements of the HSE Management and the measures resulting from it for the environment and protection of the employee's health are described in the HSE Instruction [S24]. HSE Plan is listed in Documentation Plan [DoP].

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved

APPENDIX A CRITERIA FOR THE SELECTION OF TECHNIQUES
AND MEASURES

Not applicable – solution designed according to EN50129

END OF DOCUMENT

NOT EXPORT CONTROLLED

Hitachi Rail – Internal

Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	544	16282~9950	01
3BU 17300 2001 DPAPA			

This document is not to be reproduced, modified, adapted, published, neither translated in any material form in whole or in part nor disclosed to any third party without the prior written permission of Hitachi. © 2025 Hitachi Rail Ltd, All rights reserved